

D 120206

(Pages : 2)

Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION  
MARCH 2025**

Physics/Applied Physics

PHY6B12/APH6B12—NUCLEAR PHYSICS AND PARTICLE PHYSICS

(Admissions Year—2019 Onwards)

Time : Two Hours

Maximum : 60 Marks

*The symbols used in this question paper have their usual meanings.***Section A (Short Answer Type)***Answer **all** questions in two **or** three sentences, each correct answer carries a maximum of 2 marks.*

1. Discuss any *two* properties of nuclear forces.
2. Give four evidences for the existence of magic numbers.
3. Explain the working of Scintillation Counters.
4. Mention any *four* conservation laws governing radioactive decay.
5. Why the nuclei are so small compared to the atom ?
6. Briefly give the working of Cosmotron.
7. Explain the Gamma decay process.
8. What are the basic requirements for a plasma state ?
9. What are the applications of radio isotopes ?
10. What is the strange behavior of kaons and hyperons ?
11. Explain nuclear fission.
12. Explain the terms particle and antiparticle.

(Ceiling = 20 marks)

**Turn over**

**Section B (Paragraph / Problem Type)**

Answer **all** questions in a paragraph of about **half a page to one page**, each correct answer carries a maximum of 5 marks.

13. What is binding energy ? Explain the stability of nucleus and binding energy.
14. Given the following isotopic masses  ${}_3\text{Li}^7 = 7.016004$ ,  ${}_3\text{Li}^6 = 6.015125$  and  ${}_0n^1 = 1.008665$ . Calculate the binding energy of neutron in  ${}_3\text{Li}^7$  nucleus. Express the result in  $u$ , MeV and Joules.
15. Write the working of Photographic plate ?
16. A cyclotron in which the flux density is  $1.4\text{weber/m}^2$  is employed to accelerate protons. How rapidly should the electric field between the Dees be reversed ? Mass of the proton =  $1.67 \times 10^{-27}$  kg. and charge =  $1.6 \times 10^{-19}$ .
17. Discuss the working principle of Van de Graff electrostatics generator.
18. Explain the basic features of fission reactions that make it useful as a means to generate electrical energy.
19. A reactor is developing energy at the rate of 3000 kW. How many atoms of  $\text{U}^{235}$  undergo fission per second ? How many kilograms of  $\text{U}^{235}$  would be used in 1000 hours of operation assuming that on an average energy of 200 MeV is released per fission ?

(Ceiling = 30 marks)

**Section C (Essay Type)**

Essays - Answer in about **two pages**, any **one** question.

The question carries 10 marks.

20. Explain liquid drop model of a nucleus and arrive at the semi-empirical mass formula.
21. Explain the different elementary particle quantum numbers and their conservation laws with examples.

(1 × 10 = 10 marks)